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Building a **Zero-Trust** Azure Environment

How I used CLI & Bastion to eliminate public security risks.

AZURE CLOUD

SECURITY

Secure Subnetting

By segmenting the **10.0.1.0/24** network, I isolated backend services from external threats while maintaining high availability.

Connected devices

Subnets

Monitoring

Topology

applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

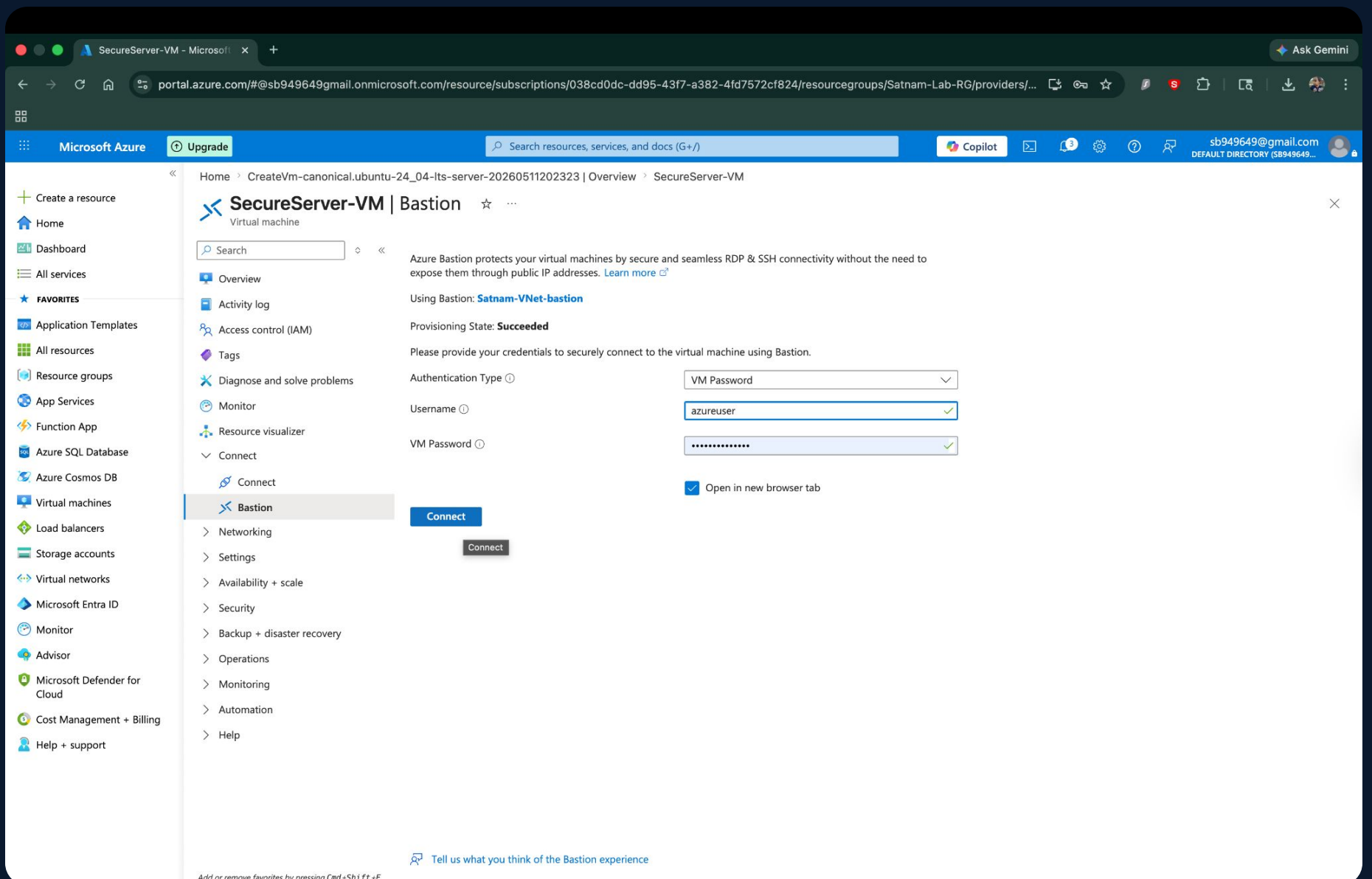
☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to
☐	BackendSu...	10.0.1.0/24	-	251	-
☐	AzureBasti...	10.0.2.0/24	-	251	-

NETWORKING

AZURE VIRTUAL NETWORK

Visibility = Risk

By removing the Public IP, I protected this server from **99%** of internet-based brute force attacks.

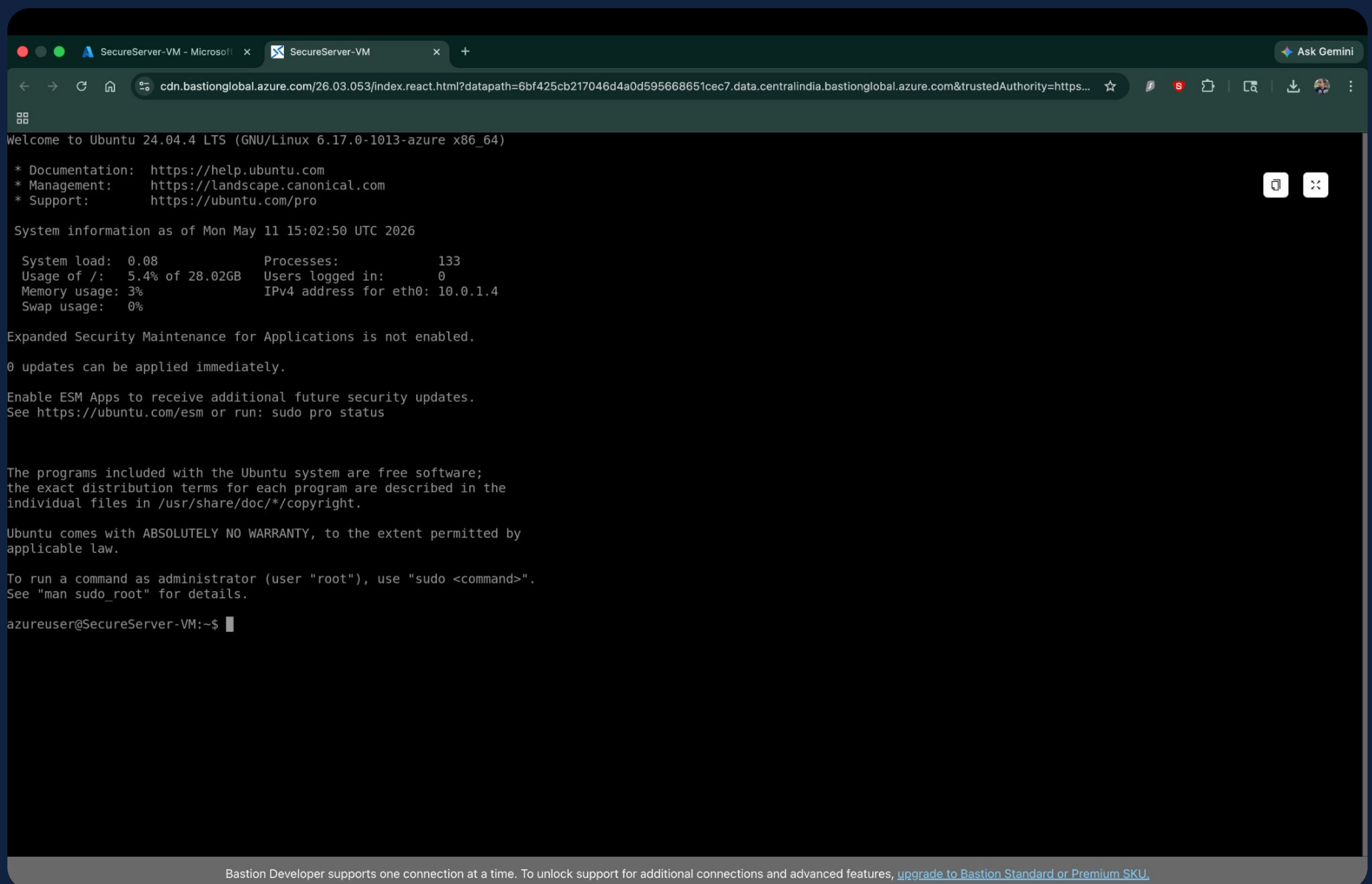


SECURITY

AZURE BASTION

Secure Management

Instead of open ports, I used an encrypted **SSL tunnel (Port 443)** to manage the infrastructure securely.



The screenshot shows a web browser window with two tabs: "SecureServer-VM - Microsoft" and "SecureServer-VM". The active tab displays a terminal window with the following content:

```
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1013-azure x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Mon May 11 15:02:50 UTC 2026

System load:  0.08          Processes:            133
Usage of /:   5.4% of 28.02GB Users logged in:        0
Memory usage: 3%           IPv4 address for eth0: 10.0.1.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@SecureServer-VM:~$
```

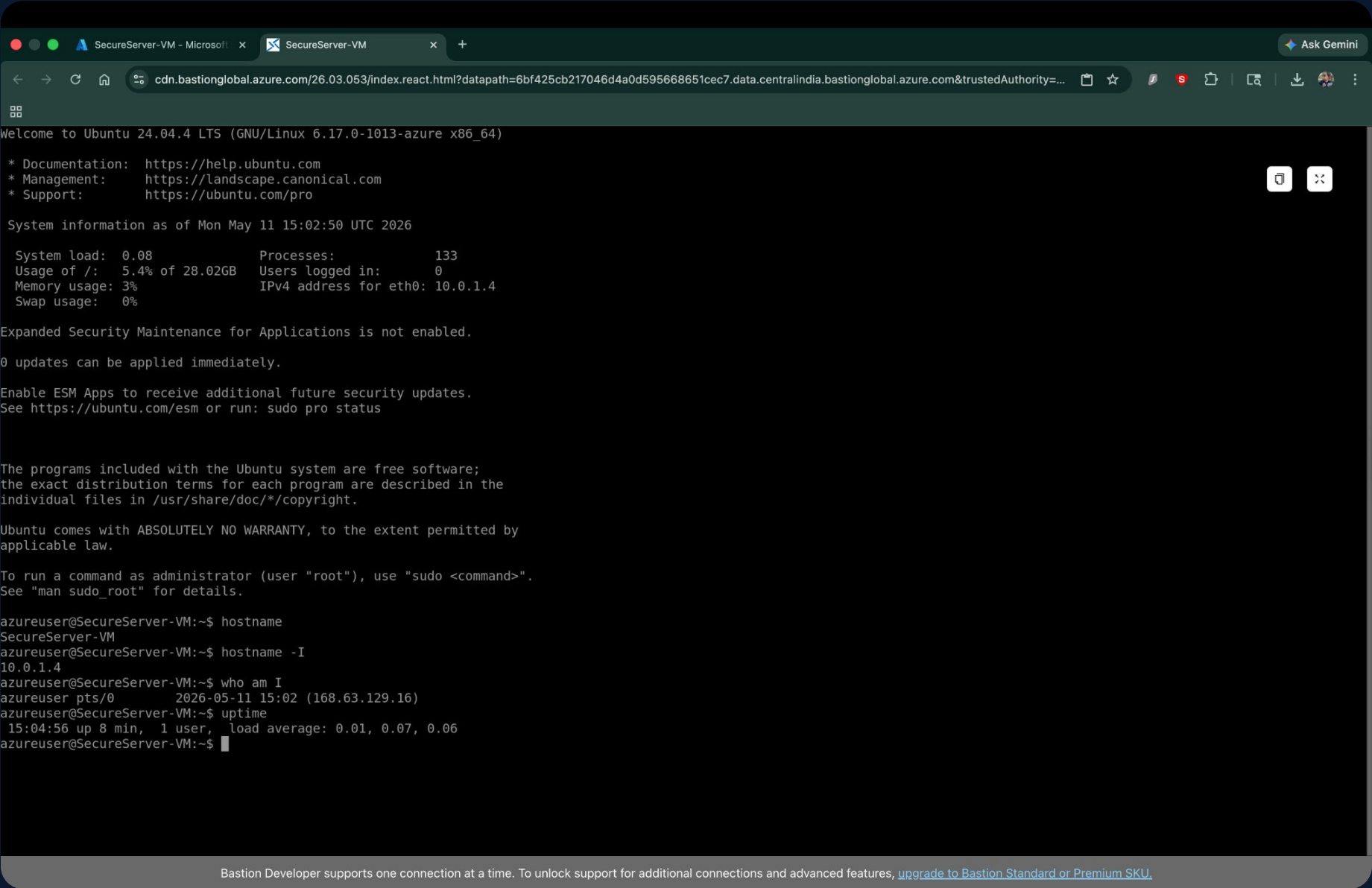
At the bottom of the browser window, there is a footer message: "Bastion Developer supports one connection at a time. To unlock support for additional connections and advanced features, [upgrade to Bastion Standard or Premium SKU.](#)"

ENCRYPTION

PORT 443

Deployment Ready

Verification: Internal IP **10.0.1.4** confirmed.
System active, secure, and ready for deployment.



VERIFICATION

NETWORK ACTIVE